Explosion Protection

Physical Fundamentals



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Overview

1. **Formation of an Explosive Atmosphere**
   Flammable Substances, Hazardous Quantity, Explosive Limits
2. **Ignition of an Explosive Atmosphere**
   Potential Sources of Ignition: Hot Surfaces, Electrical Installations, Mechanical Sparks, Static Electricity
3. **Effects from Explosions**
   Explosion Pressure, Maximum Experimental Safe Gap, Groups

1. Formation of an Explosive Atmosphere

Flammable Substances,
Hazardous Quantity,
Explosive Limits



Flammable Substances

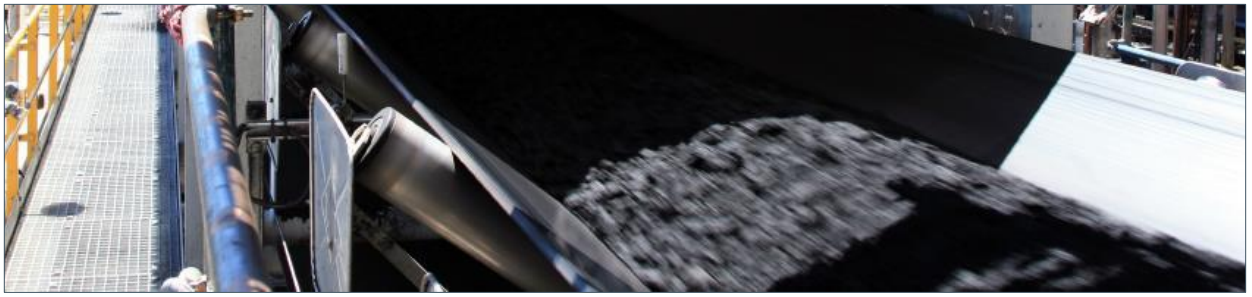


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Experience – Classification of Substances

- Flammable (e.g. coal, paper, petrol, natural gas, hydrogen)
- Non-flammable (e.g. concrete, sand, metal, water, nitrogen)
  - → *Form appearance is essential*

Classification of Equipment and Flammable Substances



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Acc. to IEC standards, both equipment as well as flammable substances are classified into groups:

- Group I: mining applications
- Group II: non-mining applications with gases or vapours (sub-groups IIA, IIB and IIC)
- Group III: non-mining applications with combustible dusts (sub-groups IIIA, IIIB and IIIC)
  - → *Criteria for sub-groups A, B and C: refer to section 2 and 3 of this presentation*

Classification in Group IIIA, IIIB und IIIC



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DUST

Group		Criterion
III	A	Combustible flyings
	B	Non conductive dust
	C	Conductive dust

Term: Combustible Flyings

Solid particles, including fibres, greater than 500 µm in nominal size which may be suspended in air and could settle out of the atmosphere under their own weight.

NOTE Examples of flyings include rayon, cotton (including cotton linters and cotton waste), sisal, jute, hemp, cocoa fibre, oakum, and baled waste kapok.

EN 60079-0:2012, 3.18.2

Term | Citation | Literature

Term: Conductive Dust

Combustible dust with electrical resistivity equal to or less than 10^3 Ω·m

EN 60079-0:2012, 3.18.1.1

Term | Citation | Literature

Explosive Limits



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Concentration of flammable substances in the air:

- Explosions are possible within substance-specific limits only (explosive limits)
- In case of dusts the concentration can significantly vary (resuspension and deposition)
  - ➝ *Deviations are possible under non-atmospheric conditions!*

Term: Explosive Limits

Lower Explosive Limit; LEL

Concentration of flammable gas or vapour in air, below which an explosive gas atmosphere will not be formed.

Upper Explosive Limit; UEL

Concentration of flammable gas or vapour in air, above which an explosive gas atmosphere will not be formed.

IEV 426-02-10 and 426-02-09

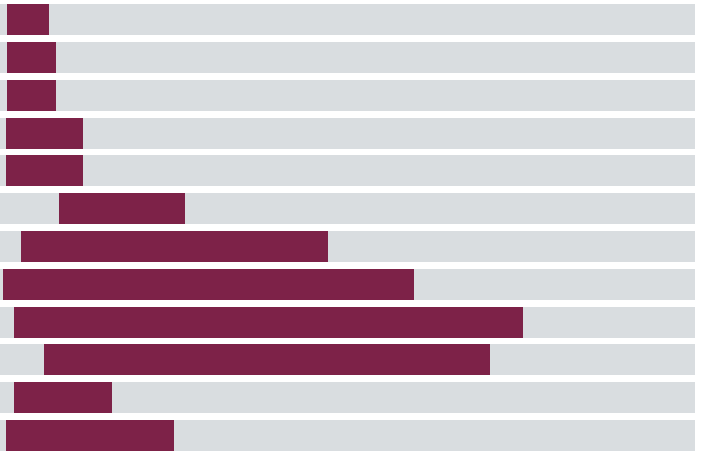
Term

Citation

Literature

Examples: Explosive Limits – Gases

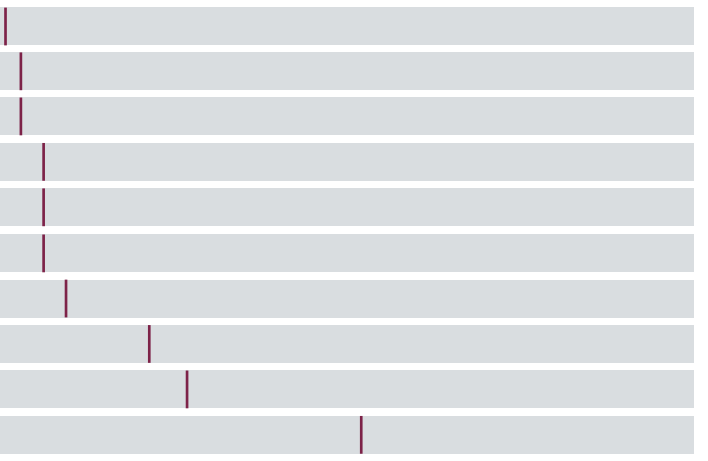
Diesel	0,6 ... 6,5 Vol.-%
Petrol	0,6 ... 8 Vol.-%
Toluol	1,1 ... 7,8 Vol.-%
Methyl Ethyl Ketone	1,5 ... 12,6 Vol.-%
Tetrahydrothiophene	1,1 ... 12,3 Vol.-%
Ammonia	15,4 ... 33,6 Vol.-%
Methanol	6 ... 50 Vol.-%
Carbon Disulphide	0,6 ... 60 Vol.-%
Hydrogen	4 ... 77 Vol.-%
Carbon Monoxide	11,3 ... 75,6 Vol.-%
Methane	4,4 ... 17 Vol.-%
Gasodor ® S-Free	1,6 ... 23 Vol.-%



Source: GESTIS (July 2013)

Examples: Lower Explosive Limits – Dusts

Toner (803)	15 g/m³
Charcoal (5671)	30 g/m³
Activated carbon from filters (5653)	30 g/m³
Skimmed milk powder(2051)	60 g/m³
Stabiliser for PVC (5810)	60 g/m³
Aluminium (672)	60 g/m³
Mustard seeds (coarsley ground) (846)	100 g/m³
Ground coffee (838)	200 g/m³
Polyamide (924)	250 g/m³
Iron powder (2466)	500 g/m³



GESTIS, August 2013 (in brackets: substance no.)

Hazardous Quantity



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Hazardous quantity of an explosive atmosphere:

- 10 ℓ of explosive atmosphere independent of the room volume
- An equivalent of 1/10.000 of the room volume in case of small rooms (i.e. 0,01 % of the volume)
  - → *Smaller quantities might be critical as well when located close to a person*

Example: Petrol

Evaporation of petrol

- Explosive limits: 0,6 … 8,0 Vol.-%
- 1 ℓ liquid petrol ≙ 196 ℓ vapour
  - → *Concentration at LEL: ≈ 32.700 ℓ explosive atmosphere*
  - → *Concentration at UEL: ≈ 2.450 ℓ explosive atmosphere*
  - → *Formation of 10 ℓ explosive atmosphere by 0,3 ml liquid petrol!*



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2. Ignition of an Explosive Atmosphere

Potential Sources of Ignition: Hot Surfaces, Electrical Installations, Mechanical Sparks, Static Electricity



Overview: Potential Sources of Ignition

- Hot surfaces
- Flames and hot gases
- Mechanical sparks
- Electrical installations
- Electric compensation currents, cathodic protection systems
- Static electricity
- Lightning
- Electromagnetic waves (10^4 - 3×10^{11} Hz)
- Electromagnetic waves (3×10^{11} Hz - 3×10^{15} Hz)
- Ionising radiation
- Ultrasonic sound
- Adiabatic compression, shock waves
- Exothermal reactions, auto-ignition of dusts

Hot Surfaces



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Hot surfaces can occur:

- With electrical (e.g. luminaire) and mechanical equipment (e.g. Pump)
- In normal mode of operation (e.g. heat transfer oil, trace heating, steam pipe)
- Under fault conditions (e.g. bearing, motor winding)
- During maintenance work (e.g. welding, drilling, grinding)
- As a result of insufficient lubrication (e.g. bearing) or monitoring (e.g. dry run)

Hot Surfaces



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Ignition results from:

- Surface temperature exceeds the ignition temperature (gases, vapours, dust clouds)
- Formation of smouldering nest in combustible dust layers
  - → *Classification of flammable gases and vapours into temperature classes according to their ignition temperature*

Term: Ignition Temperature

Ignition temperature of an explosive gas atmosphere

Lowest temperature of a heated surface which, under specified conditions, will ignite a flammable substance in the form of a gas or vapour mixture with air.

NOTE The specified conditions are given in IEC 60079-4.

IEV 426-02-01

Term

Citation

Literature

Examples: Ignition Temperatures – Gases



Temperature Classes – Gas



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Flammable gases and vapours are classified to temperature classes:

- Assignment depends on the specific ignition temperature
- Classification in T1 … T6
  - → *No classification of combustible dusts into temperature classes!*

Examples: Temperature Classes

Temperature Class (Ignition Temperature > ... °C)					
T1 (> 450 °C)	**T2** (> 300 °C)	**T3** (> 200 °C)	**T4** (> 135 °C)	**T5** (> 100 °C)	**T6** (> 85 °C)
Acetone	Ethanol	Hexane	Acetaldehyde		Carbon Disulphide
Acetic Acid	Cyclohexanone	Diesel	Diethyl Ether		
Methane	Methanol	Petrol			
Propane	Ethene	Domestic Fuel Oil			
Ammonia	Butane	Hydrogen Sulphide			
Benzol	Acetylene				
Toluol					
Hydrogen Cyanide					
Hydrogen					

Temperature Classes and Equipment



Peter Leitenmaier, TÜV SÜD Industrie Service



Peter Leitenmaier, TÜV SÜD Industrie Service

Example: solvent storage

- A flammable substance is characterised by a temperature class, e.g. T4
  - → *Ignition temperature > 135 °C*

Example: Luminaire

- Equipment is characterised by a maximum surface temperature, e.g. T4
  - → *Surface temperature < 135 °C*

Characteristics of Combustible Dusts



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Small dust particles appear as:

- A dispersed dust cloud
- A deposited dust layer
  - → *Form of appearance and ignition temperature depend on the particle size*

Ignition Temperature of a Dust Layer

The lowest temperature of a hot surface at which ignition occurs in a dust layer of specified thickness on this hot surface.

Note: often referred to as “T_{5mm}” in case of layer of 5 mm thickness

EN 60079-0:2012, 3.39

Term

Citation

Literature

Ignition Temperature of a Dust Cloud

The lowest temperature of a hot inner wall of a furnace at which ignition occurs in a dust cloud in air contained therein.

Note: partly referred to as “T_{CL}”

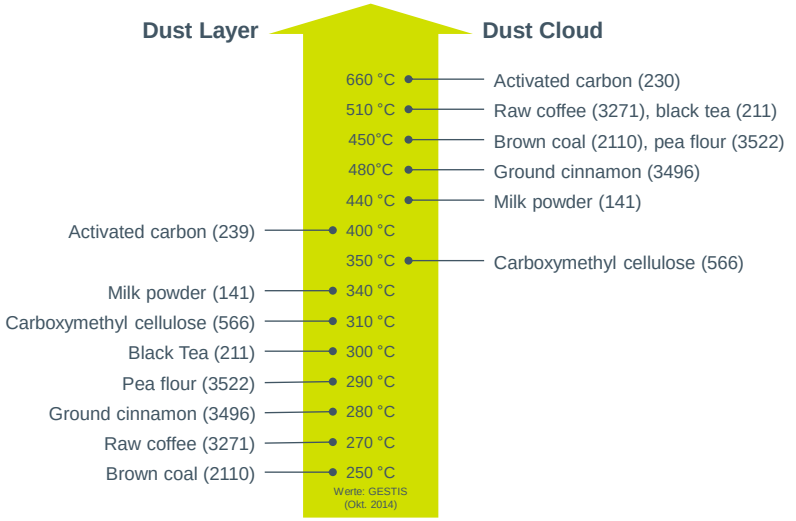
EN 60079-0:2012, 3.38

Term

Citation

Literature

Examples: Ignition Temperatures – Dusts



Surface Temperatures and Equipment



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Example: packaging of flour

- In case of combustible dusts two ignition temperatures have to be considered
  - → *E.g. T_{5mm} = 300 °C, T_{CL} = 460 °C*

Example: Luminaire

- Equipment for dust areas are marked with the actual surface temperature
  - → *Safety margins have to be considered!*

Electrical Installations

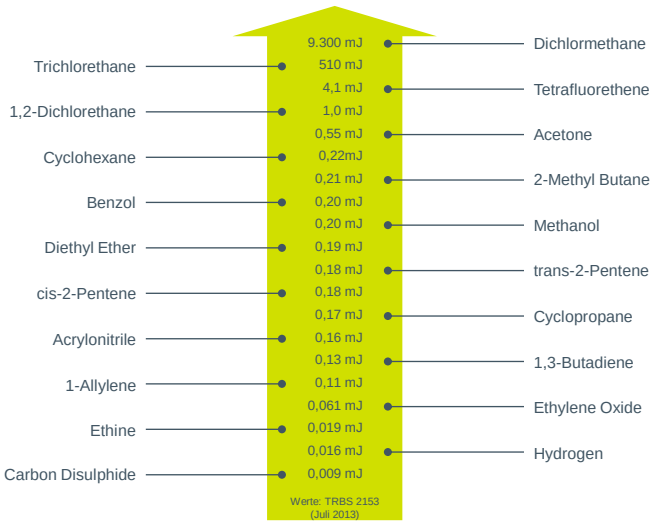


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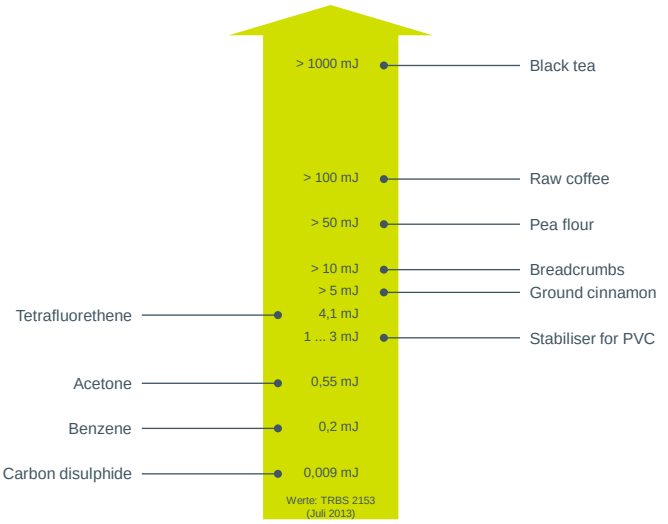
Ignition results from:

- Formation of impermissibly hot surfaces
- Generation of ignitable electrical sparks
  - ➝ *Spark ignition: classification of flammable gases and vapours into (explosion-, gas-) groups according to their ignition energy*

Examples: Ignition Energies – Gases



Comparison: Gases and Dusts



Examples: Groups and Ignition Energies

Group	Temperature Class T1 > 450 °C	T2 > 300 °C	T3 > 200 °C	T4 > 135 °C	T5 > 100 °C	T6 > 85 °C
IIA	Acetone Acetic Acid Methane Propane Ammonia Benzol Toluol	Cyclo-hexanone Methanol Butane	Hexane Diesel Petrol Domestic Fuel Oil	Acetaldehyde		
IIB	Hydrogen-Cyanide	Ethanol Ethene	Hydrogen-Sulphide	Diethyl Ether		
IIC	Hydrogen	Acetylene				Carbon-Disulphide

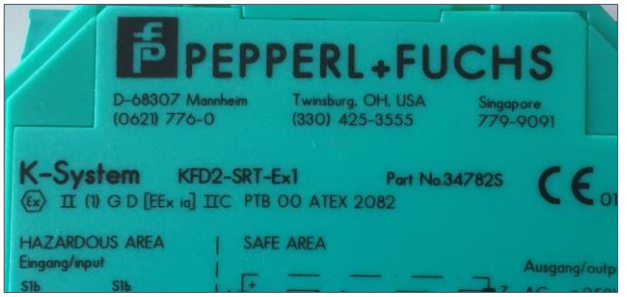
Increasing ignition energy

Werte: GESTIS (Juli 2013)

Ignition Energies and Equipment



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Example: Power Plant

- Generator cooling with hydrogen, i.e. flammable substance of group IIC
  - → *Substance requires a very low ignition energy*

Example: intrinsically safe barrier

- Barrier limits voltage current, power
  - → *Spark energy is not sufficient to ignite IIC-substances*

Mechanically Generated Sparks



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Mechanical sparks can be generated:

- When foreign particles enter moving parts (e.g. stone in a mill, screw in a ventilator)
- During maintenance work (e.g. application of tools)
  - → *Increased risk of ignition in case of oxidisable substances or showers of sparks*

Static Electricity



BG RCI

Static Electricity can occur:

- Due to the operation of plants (e.g. foils running over rolls, liquids in pipes)
- During cleaning work (e.g. wiping surfaces)
- Walking
- Etc.

3. Effects from Explosions

Explosion Pressure, Maximum Experimental Safe Gap, Groups



Characteristic Values of Explosions

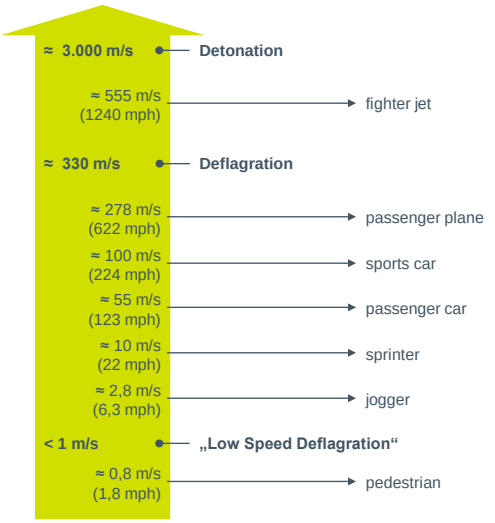


Chiltern Air Support Unit and Hertfordshire County Council

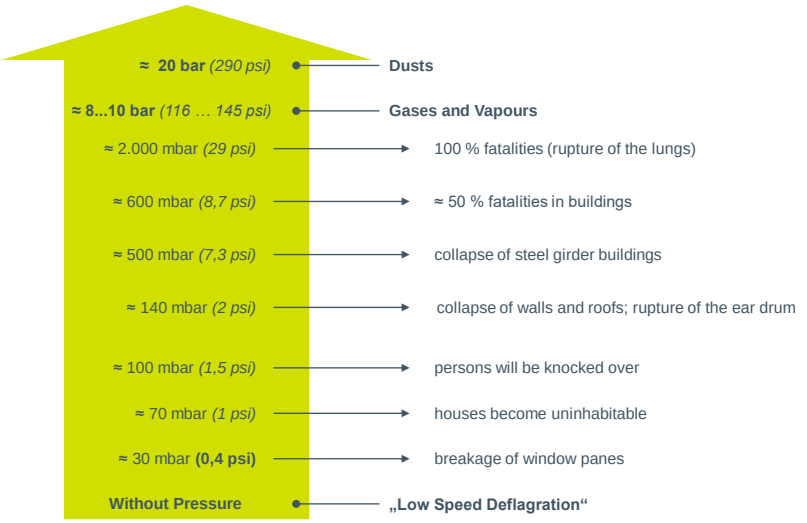
In suitable test chambers characteristic values can be observed:

- Propagation speed
- Explosion pressure
- Rise rate of the pressure
- Propagation through small gaps

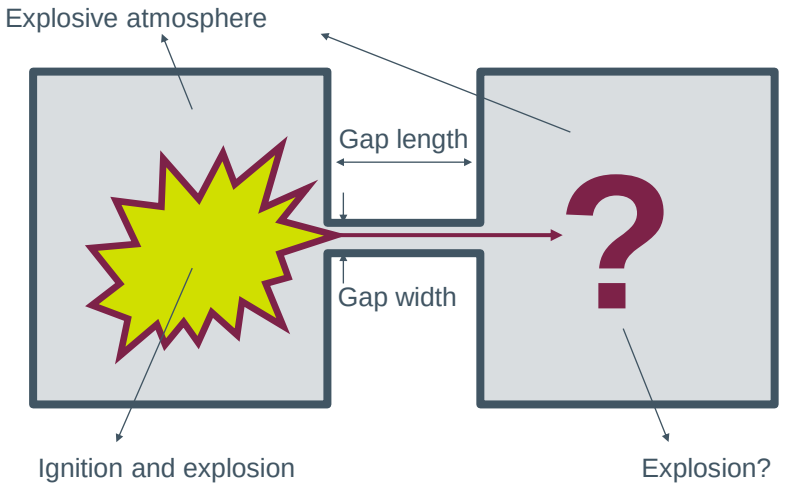
Examples: Propagation Speed



Examples: Explosion Pressure



Explosion Propagation through Gaps



Gap Width and Groups

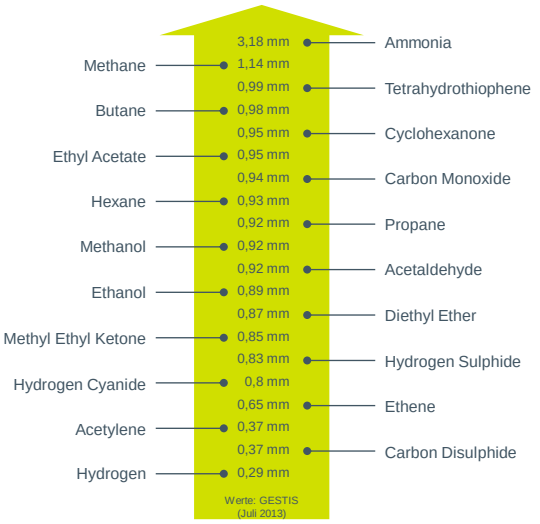


Chiltern Air Support Unit and Hertfordshire County Council

Observation during ignition experiments with different substances:

- Gas and vapour explosions propagate through substance-specific gaps
  - → *Classification of flammable gases and vapours in (explosion, gas) groups according to their Maximum Experimental Safe Gap (MESG)*

Examples: MESG



Examples: Groups and Gap Dimensions

Group	Temperature Class T1 > 450 °C	T2 > 300 °C	T3 > 200 °C	T4 > 135 °C	T5 > 100 °C	T6 > 85 °C
IIA	Acetone Acetic Acid Methane Propane Ammonia Benzol Toluol	Cyclo-hexanone Methanol Butane	Hexane Diesel Petrol Domestic Fuel Oil	Acetaldehyde		
IIB	Hydrogen-Cyanide	Ethanol Ethene	Hydrogen-Sulphide	Diethyl Ether		
IIC	Hydrogen	Acetylene				Carbon-Disulphide

Increasing gap width

Classification in Group IIA, IIB und IIC



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GAS

Group		Criterion
II	A	high ignition energy/wide MESG
	B	medium ignition energy/MESG
	C	low ignition energy/small MESG

MESG and Equipment



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Example: Solvent Storage

- Solvent is characterised by MESG, e.g. IIC
  - → *Hot gases and flames propagate through a very narrow gap*

Example: flameproof luminaire

- Equipment is characterised by its joints, e.g. IIC
  - → *Internal explosions cannot propagate to the surrounding atmosphere*